

DEMATEL MCDM Analysis Report

Native fuzzy DEMATEL preserves triangular/trapezoidal expert judgments through component-wise direct, normalized, and total-relation matrices, then defuzzifies D/R indicators for cause-effect grouping.

Study Summary

Alternatives/factors: 7. Criteria/factors: 7. Top result: CO2 Emissions.

94 bundled numerical checks and 43 external published-example fixtures are included; TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Certification status: not publication-certified for every variant. Only registered passing external fixtures count as published-example validation evidence for the selected method and variant.

Study Mode Evidence

Input type: Crisp numeric input. No triangular/trapezoidal fuzzy cells were detected in this run.

Respondent/group aggregation: No respondent matrix aggregation used. The uploaded/active decision matrix was treated as a single already aggregated dataset.

Expert aggregation: 2 expert matrixes; Strong consensus. Mean disagreement: 0; relative disagreement: 0.

Fuzzy calculation convention: Component-wise fuzzy total relation. DEMATEL reports whether fuzzy tuples stayed component-wise through total relation or were defuzzified before the final causal matrix.

Validation Evidence

Certification status: Not publication-certified for every variant. Treat internal checks as implementation evidence. Treat only registered passing external fixtures as published-example validation for the selected method and variant.

Selected method validation status: Validated with discrepancies tracked. 1 passing published-example fixture; 1 validation candidate tracked separately

Internal workflow and algorithm checks: Included. Registry integrity, method execution, group data, fuzzy input, zero-value handling, validation rules, templates, reports, and coverage summaries.

Bundled numerical benchmark checks: 94 checks. Known expected rankings and key coefficients for the built-in sample suites.

Selected method benchmark coverage: DEMATEL. Selected method has bundled numerical benchmark coverage

External published-example validation: In progress: 43 fixtures. TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Selected method external validation: DEMATEL. 1 external published-example fixture registered for this selected method: crisp-average-expert-direct-matrix-mean-threshold; 1 validation candidate also tracked.

External fixture: DEMATEL crisp-average-expert-direct-matrix-mean-threshold. Sage Open 2025 collaborative innovation DEMATEL example; DOI 10.1177/21582440251387390. DEMATEL average expert direct-relation matrix, row/column normalization factor, total-relation matrix, mean-threshold relation matrix, D/R cause-effect indicators, and prominence ranking. Source: <https://journals.sagepub.com/doi/10.1177/21582440251387390>

External fixture: DEMATEL native-fuzzy-triangular-smart-manufacturing (discrepancy candidate). Sustainability 2023 smart-manufacturing fuzzy DEMATEL example; DOI 10.3390/su15043864. Published triangular fuzzy DEMATEL input and final cause/effect table are tracked, but the final D/R magnitudes appear to use a defuzzified or scaled convention different from the app native fuzzy component-wise pipeline. Source: <https://www.mdpi.com/2071-1050/15/4/3864>

Calculation Steps

1. Normalize direct relation matrix.

- 2. Compute total relation matrix $T = N(I-N)^{-1}$.
- 3. Calculate D, R, D+R, and D-R.
- 4. Classify factors into cause and effect groups.

Analyzed Direct Relation Matrix

Source factor	C1	C2	C3	C4	C5	C6	C7
Cost	0	2	3	4	1	2	3
Environmental Management	2	0	4	1	2	3	4
CO2 Emissions	3	4	0	2	3	4	1
Labor Practices	4	1	2	0	4	1	2
Innovation	1	2	3	4	0	2	3
Community Impact	2	3	4	1	2	0	4
Waste	3	4	1	2	3	4	0

Fuzzy Direct Relation Matrix

Factor	C1	C2	C3	C4	C5	C6	C7
Cost	(0, 0, 0)	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)	(3, 3, 3)
Environmental Management	(2, 2, 2)	(0, 0, 0)	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)
CO2 Emissions	(3, 3, 3)	(4, 4, 4)	(0, 0, 0)	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)	(1, 1, 1)
Labor Practices	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)	(0, 0, 0)	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)
Innovation	(1, 1, 1)	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)	(0, 0, 0)	(2, 2, 2)	(3, 3, 3)
Community Impact	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)	(0, 0, 0)	(4, 4, 4)
Waste	(3, 3, 3)	(4, 4, 4)	(1, 1, 1)	(2, 2, 2)	(3, 3, 3)	(4, 4, 4)	(0, 0, 0)

Fuzzy Expert Aggregation Summary

Expert matrices	Aggregation	Mean absolute disagreement	Max absolute disagreement	Relative disagreement	Consensus level	Fuzzy tuple aggregation
2	Arithmetic mean	0	0	0	Strong consensus	Fuzzy tuple components aggregated by arithmetic mean

Fuzzy Normalized Direct Relation Matrix

Factor	C1	C2	C3	C4	C5	C6	C7
Cost	(0, 0, 0)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)
Environmental Management	(0.1176, 0.1176, 0.1176)	(0, 0, 0)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)
CO2 Emissions	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0, 0, 0)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)
Labor Practices	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0, 0, 0)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)
Innovation	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0, 0, 0)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)
Community Impact	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0, 0, 0)	(0.2353, 0.2353, 0.2353)
Waste	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0.0588, 0.0588, 0.0588)	(0.1176, 0.1176, 0.1176)	(0.1765, 0.1765, 0.1765)	(0.2353, 0.2353, 0.2353)	(0, 0, 0)

Fuzzy Total Relation Matrix

Factor	C1	C2	C3	C4	C5	C6	C7
Cost	(1.5804, 1.5804, 1.5804)	(1.8061, 1.8061, 1.8061)	(1.9001, 1.9001, 1.9001)	(1.6517, 1.6517, 1.6517)	(1.636, 1.636, 1.636)	(1.8061, 1.8061, 1.8061)	(1.9001, 1.9001, 1.9001)
Environmental Management	(1.8061, 1.8061, 1.8061)	(1.8656, 1.8656, 1.8656)	(2.1015, 2.1015, 2.1015)	(1.6315, 1.6315, 1.6315)	(1.8061, 1.8061, 1.8061)	(2.0156, 2.0156, 2.0156)	(2.1015, 2.1015, 2.1015)
CO2 Emissions	(1.9001, 1.9001, 1.9001)	(2.1015, 2.1015, 2.1015)	(1.9833, 1.9833, 1.9833)	(1.7322, 1.7322, 1.7322)	(1.9001, 1.9001, 1.9001)	(2.1015, 2.1015, 2.1015)	(2.0388, 2.0388, 2.0388)
Labor Practices	(1.6517, 1.6517, 1.6517)	(1.6315, 1.6315, 1.6315)	(1.7322, 1.7322, 1.7322)	(1.3768, 1.3768, 1.3768)	(1.6517, 1.6517, 1.6517)	(1.6315, 1.6315, 1.6315)	(1.7322, 1.7322, 1.7322)
Innovation	(1.636, 1.636, 1.636)	(1.8061, 1.8061, 1.8061)	(1.9001, 1.9001, 1.9001)	(1.6517, 1.6517, 1.6517)	(1.5804, 1.5804, 1.5804)	(1.8061, 1.8061, 1.8061)	(1.9001, 1.9001, 1.9001)
Community Impact	(1.8061, 1.8061, 1.8061)	(2.0156, 2.0156, 2.0156)	(2.1015, 2.1015, 2.1015)	(1.6315, 1.6315, 1.6315)	(1.8061, 1.8061, 1.8061)	(1.8656, 1.8656, 1.8656)	(2.1015, 2.1015, 2.1015)
Waste	(1.9001, 1.9001, 1.9001)	(2.1015, 2.1015, 2.1015)	(2.0388, 2.0388, 2.0388)	(1.7322, 1.7322, 1.7322)	(1.9001, 1.9001, 1.9001)	(1.9001, 1.9001, 1.9001)	(1.9001, 1.9001, 1.9001)

(2.1015, 2.1015, 2.1015) | (1.9833, 1.9833, 1.9833)

Fuzzy DEMATEL Thresholded Total Relation Matrix

Alternative		C1		C2		C3		C4		C5		C6		C7
Cost		0		0		1.9001		0		0		0		1.9001
Environmental Management		0		0		2.1015		0		0		2.0156		2.1015
CO2 Emissions		1.9001		2.1015		0		0		1.9001		2.1015		2.0388
Labor Practices		0		0		0		0		0		0		0
Innovation		0		0		1.9001		0		0		0		1.9001
Community Impact		0		2.0156		2.1015		0		0		0		2.1015
Waste		1.9001		2.1015		2.0388		0		1.9001		2.1015		0

Fuzzy DEMATEL Cause-Effect Table

Factor		Name		Fuzzy D		Fuzzy R		D+R centroid		D-R centroid		Group
C1		Cost		(12.2804, 12.2804, 12.2804)		(12.2804, 12.2804, 12.2804)		24.5608		0		Cause
C2		Environmental Management		(13.3278, 13.3278, 13.3278)		(13.3278, 13.3278, 13.3278)		26.6556		0		Cause
C3		CO2 Emissions		(13.7575, 13.7575, 13.7575)		(13.7575, 13.7575, 13.7575)		27.515		0		Cause
C4		Labor Practices		(11.4076, 11.4076, 11.4076)		(11.4076, 11.4076, 11.4076)		22.8152		0		Effect
C5		Innovation		(12.2804, 12.2804, 12.2804)		(12.2804, 12.2804, 12.2804)		24.5608		0		Cause
C6		Community Impact		(13.3278, 13.3278, 13.3278)		(13.3278, 13.3278, 13.3278)		26.6556		0		Effect
C7		Waste		(13.7575, 13.7575, 13.7575)		(13.7575, 13.7575, 13.7575)		27.515		0		Cause

Figure data: Fuzzy Cause-Effect Scatter

factor		prominence		relation		group
C1		24.5608		0		Cause
C2		26.6556		0		Cause
C3		27.515		0		Cause
C4		22.8152		0		Effect
C5		24.5608		0		Cause
C6		26.6556		0		Effect
C7		27.515		0		Cause

Figure data: Fuzzy Total Relation Centroid Heatmap

source		target		value
C1		C1		1.5804
C1		C2		1.8061
C1		C3		1.9001
C1		C4		1.6517
C1		C5		1.636
C1		C6		1.8061
C1		C7		1.9001
C2		C1		1.8061
C2		C2		1.8656
C2		C3		2.1015
C2		C4		1.6315
C2		C5		1.8061
C2		C6		2.0156
C2		C7		2.1015
C3		C1		1.9001
C3		C2		2.1015
C3		C3		1.9833
C3		C4		1.7322
C3		C5		1.9001
C3		C6		2.1015
C3		C7		2.0388
C4		C1		1.6517
C4		C2		1.6315
C4		C3		1.7322
C4		C4		1.3768
C4		C5		1.6517
C4		C6		1.6315
C4		C7		1.7322

C5 | C1 | 1.636
C5 | C2 | 1.8061
C5 | C3 | 1.9001
C5 | C4 | 1.6517
C5 | C5 | 1.5804
C5 | C6 | 1.8061
C5 | C7 | 1.9001
C6 | C1 | 1.8061
C6 | C2 | 2.0156
C6 | C3 | 2.1015
C6 | C4 | 1.6315
C6 | C5 | 1.8061
C6 | C6 | 1.8656
C6 | C7 | 2.1015
C7 | C1 | 1.9001
C7 | C2 | 2.1015
C7 | C3 | 2.0388
C7 | C4 | 1.7322
C7 | C5 | 1.9001
C7 | C6 | 2.1015
C7 | C7 | 1.9833